Introduction to R

https://tinyurl.com/hbc-r-flipped

Harvard Chan Bioinformatics Core

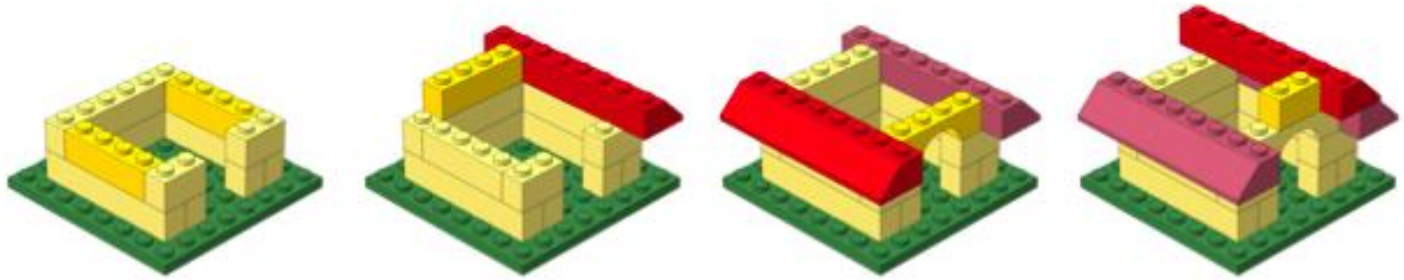
Learning Objectives

- Comfortably use RStudio (a graphical interface for R)
- Fluently interact with R using RStudio
- Become familiar with R syntax
- Understand data structures in R
- Inspect and manipulate data structures
- Install packages and use functions in R
- Visualize data using ggplot2
- Utilize pipes, tibbles and functions from the Tidyverse package suite

Exit survey

https://tinyurl.com/r-workshop-hbc

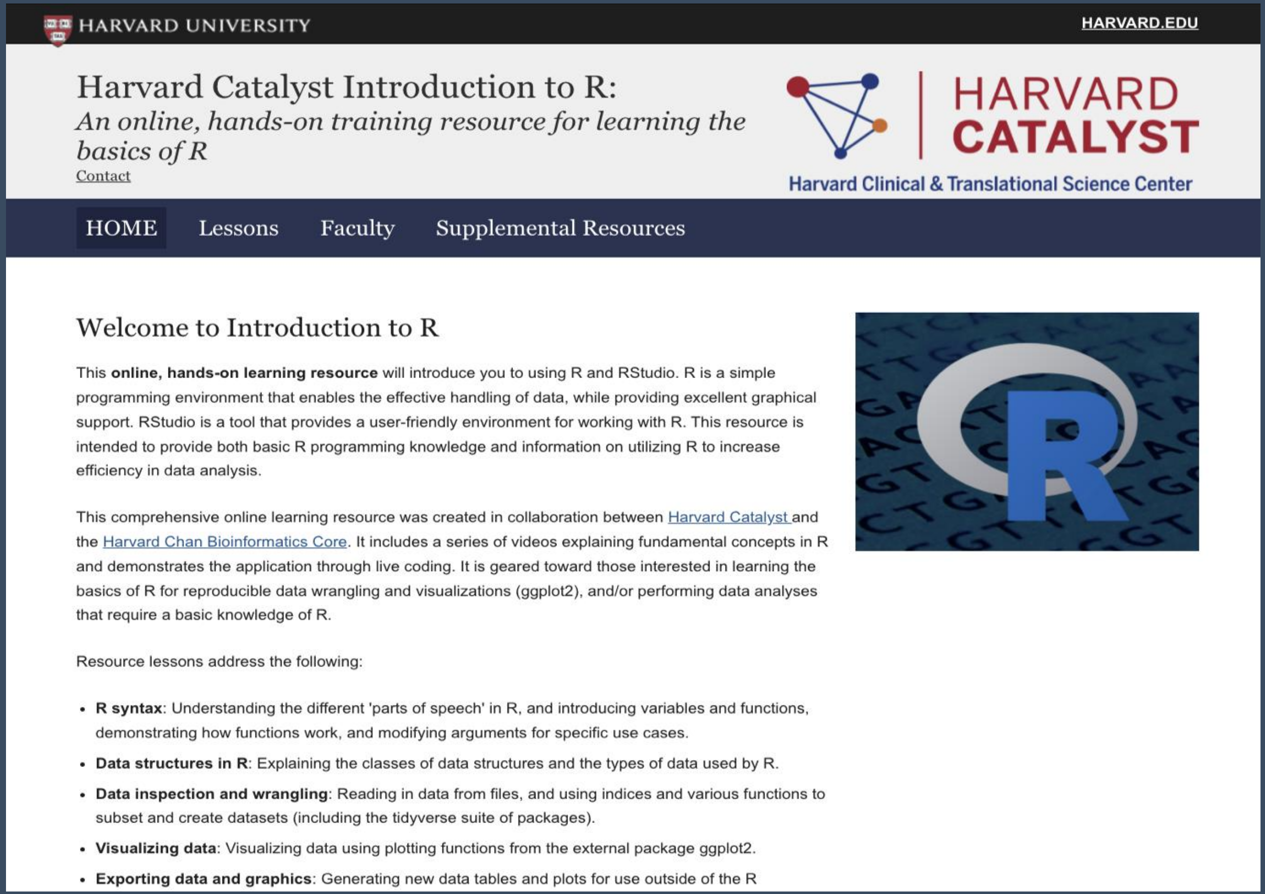
Keep building!



Module	Pre-requisite	Date	Time	Registration Page
The cBioPortal for Cancer Genomics	None	9/23/26	9:30am-12:30pm	Register here
Single Cell RNA-seq Analysis Using Cellenics	HarvardKey Credentials	10/21/26	9:30am-12:30pm	Register here
Multiomics Part I*	Foundations in R	11/11/26	9:30am-12:30pm	TBD
Multiomics Part II*	Multiomics Part I	12/16/26	9:30am-12:30pm	TBD
Multiomics Part III*	Multiomics Part II	1/20/27	9:30am-12:30pm	TBD

https://hsph.harvard.edu/research/bioinformatics/training/training-schedule/

Harvard Catalyst Online Resource



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Harvard Catalyst Introduction to R:

An online, hands-on training resource for learning the basics of R

Contact

HARVARD CATALYST

Harvard Clinical & Translational Science Center

HOME Lessons Faculty Supplemental Resources

Welcome to Introduction to R

This **online, hands-on learning resource** will introduce you to using R and RStudio. R is a simple programming environment that enables the effective handling of data, while providing excellent graphical support. RStudio is a tool that provides a user-friendly environment for working with R. This resource is intended to provide both basic R programming knowledge and information on utilizing R to increase efficiency in data analysis.

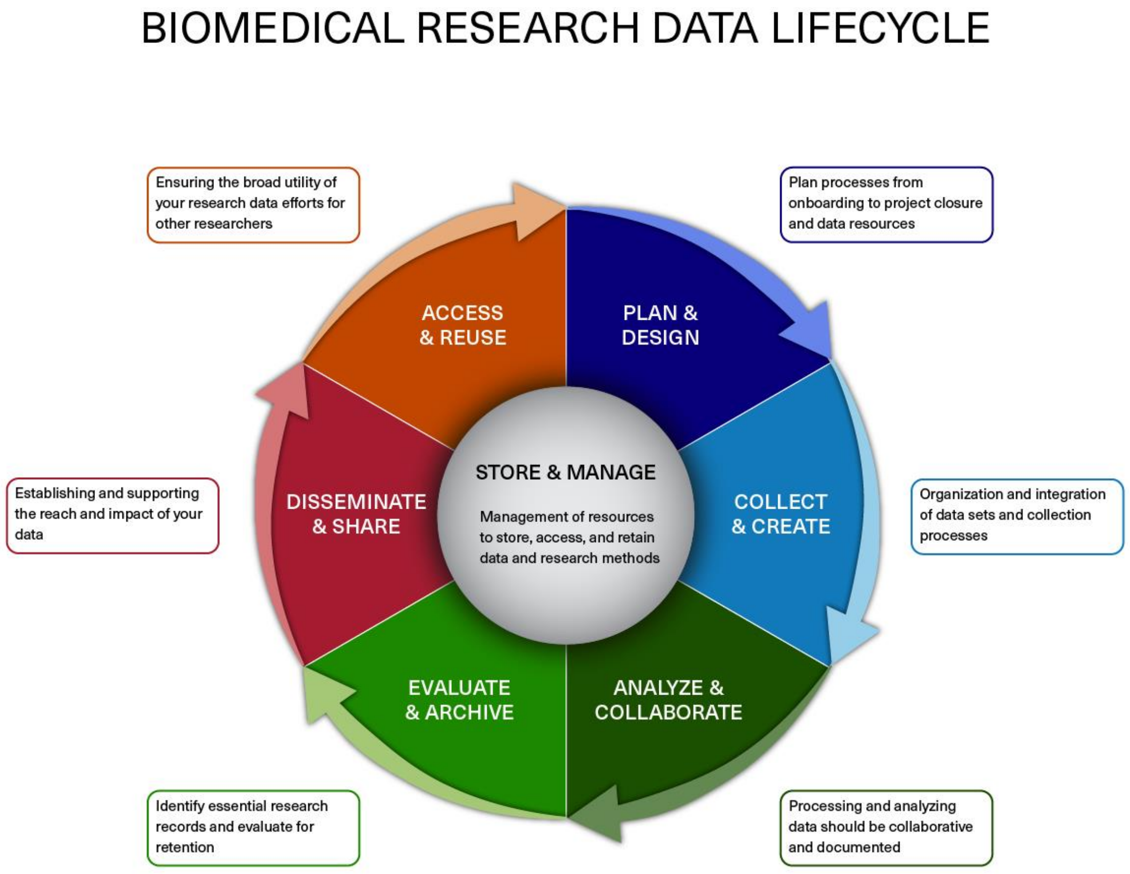
This comprehensive online learning resource was created in collaboration between Harvard Catalyst and the Harvard Chan Bioinformatics Core. It includes a series of videos explaining fundamental concepts in R and demonstrates the application through live coding. It is geared toward those interested in learning the basics of R for reproducible data wrangling and visualizations (ggplot2), and/or performing data analyses that require a basic knowledge of R.

Resource lessons address the following:

- **R syntax**: Understanding the different 'parts of speech' in R, and introducing variables and functions, demonstrating how functions work, and modifying arguments for specific use cases.
- **Data structures in R**: Explaining the classes of data structures and the types of data used by R.
- **Data inspection and wrangling**: Reading in data from files, and using indices and various functions to subset and create datasets (including the tidyverse suite of packages).
- **Visualizing data**: Visualizing data using plotting functions from the external package ggplot2.
- **Exporting data and graphics**: Generating new data tables and plots for use outside of the R

https://ondemand.catalyst.harvard.edu/products/Introduction-to-R

Research Data Management (RDM)



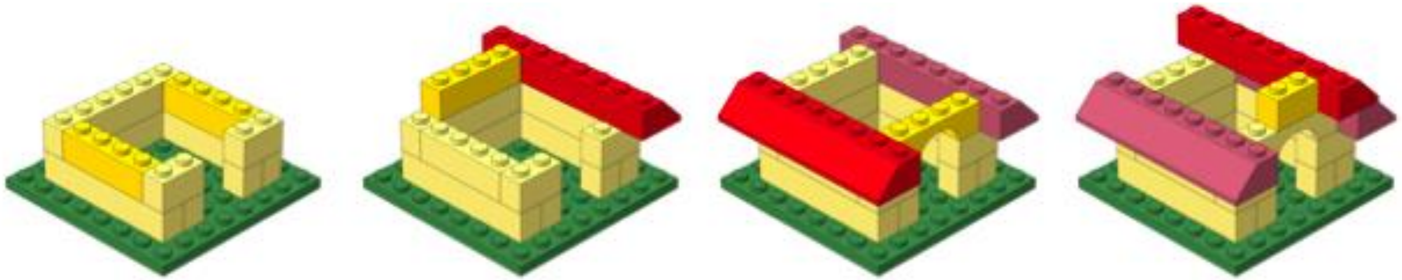
Better RDM practice benefits you

- **HMS Data Management LMA**
  - **Webpage:** https://datamanagement.hms.harvard.edu
  - **Sign up for quarterly email updates**
- **Harvard-wide Research data Management**
  - https://researchdatamanagement.harvard.edu/

Date	Time	Event	Location
Sep 11	9am	**HKS Data Carpentry Day 1: Tidy Data, OpenRefine, and Intro to R**	HKS Wexner Building
Sep 14	2pm	**protocols.io Webinar: Tips and Tricks**	Zoom
Sep 15	10am	**Harvard Dataverse: HEAL Data Package (HDP) Tool Demo**	Zoom
Sep 16	12pm	**Data Discussions: Let's Talk Data Services**	Zoom
Sep 17	12pm	**Organizing and Archiving Qualitative Data with the Qualitative Data Repository (QDR)**	
Sep 22	12pm	**Biomedical and Public Health Research Data Management: Concepts, Tools, and Strategies**	Countway Library, Classroom L1-028
Sep 23	9:30am	**HBC: The cBioPortal for Cancer Genomics**	Zoom
Sep 23	12pm	**Research Data Management: Concepts, Tools, and Strategies**	Zoom
Sep 24	11am	**Registration Templates on the OSF**	Zoom
Sep 24	12pm	**Scientific Research Data Management: Concepts, Tools, and Strategies**	Cabot Library, Instruction Room LL-03

https://datamanagement.hms.harvard.edu/training-events/rdmwg-calendar

Keep building!



Workshop	Dates	Time	Location	Registration Page*
Introduction to Python	November 10, 13, 17 & 20, 2026	10am-12pm	Zoom	Register here
Introduction to R	May 4, 7, 11 & 14, 2027	10am-12pm	Zoom	TBD

Workshop	Pre-requisite*	Dates	Time	Location	Registration Page**
Introduction to Spatial Transcriptomics	Introduction to R	October 6, 9, 13 & 16, 2026	10am-12pm	Zoom	Register here
Tools for Reproducible Research	Introduction to R	October 27, 30 & November 3, 2026	9:30am-12pm	Zoom	Register here
Introduction to Spatial Transcriptomics	Introduction to R	January 26, 29, February 2 & 5, 2027	10am-12pm	Zoom	TBD
Introduction to scRNA-seq (R-based)	Introduction to R	February 16, 19, 23 & 26, 2027	10am-12pm	Zoom	Register here
Pseudobulk and related approaches for scRNA-seq analysis	Introduction to R	March 16, 19, 23 & 26, 2027	10am-12pm	Zoom	Register here
Introduction to Differential Gene Expression Analysis	Introduction to R	May 25, 28, June 1 & 4, 2027	10am-12pm	Zoom	Register here

https://hsph.harvard.edu/research/bioinformatics/training/training-schedule/

Talk to us early!

Involvement in study design to optimize experiments

More Information

- *HBC training materials: https://hbctraining.github.io/main*
- *HBC website: https://hsph.harvard.edu/research/bioinformatics/*

Contact Us

Sign up for our mailing list:

https://tinyurl.com/hbc-training-mailing-list

- *HBC training team:* hbctraining@hsph.harvard.edu
- *HBC consulting:* bioinformatics@hsph.harvard.edu